

Notes: Campello, Lin, Ma & Zou (JF, 2011)
The Real and Financial Implications of Corporate Hedging

Contents

1	Big picture	2
2	Incremental contribution	2
3	Data and measurement	2
4	Models and empirical specifications	2
4.1	Loan spread model	2
4.2	Capital expenditure restrictions	2
4.3	Investment model	2
4.4	True cost of debt	2
5	Identification and empirical design	3
6	Tables	3
6.1	Table I: Summary Statistics	3
6.2	Tables II and III: Loan pricing and capex covenants	4
6.3	Tables IV and VI: Investment and true cost of debt	4
6.4	Table V: Robustness Tests	5
7	Result map	5
8	Conclusion	7

1 Big picture

- **Question.** Does corporate hedging affect real corporate outcomes, or is it merely a financial side policy?
- **Core mechanism.** Hedging reduces left-tail cash-flow realizations and limits risk shifting, so lenders should charge lower spreads, impose fewer investment restrictions, and allow greater investment.
- **Main evidence.** Hedging lowers loan spreads, reduces capital expenditure restrictions, raises investment, and lowers estimated true cost of debt.

2 Incremental contribution

- Moves hedging research from firm-value correlations to direct financial-contracting mechanisms.
- Studies creditors' evaluation of hedging rather than only shareholders' valuation of hedging.
- Hand-collects derivatives use and links it to private loan pricing and covenants.
- Uses tax convexity as an instrument for hedging intensity.
- Shows two precise channels: lower debt costs and fewer investment restrictions.
- Links financial contracting to real capital expenditure.

3 Data and measurement

The paper hand-collects IR and FX derivatives use from SEC filings and matches it to syndicated loan contracts. Key variables include IR/FX hedging intensity, loan spreads, capital expenditure restrictions, investment scaled by lagged assets, and true cost of debt. Controls include size, profitability, tangibility, leverage, Z-score, M/B, asset volatility, loan size, maturity, loan purpose, rating dummies, and macro spreads.

4 Models and empirical specifications

4.1 Loan spread model

$$\log(\text{LoanSpread}) = f(\text{Hedging}, \text{FirmCharacteristics}, \text{LoanCharacteristics}, \text{MacroControls}, \text{FixedEffects}).$$

4.2 Capital expenditure restrictions

$$\Pr(\text{CAPEXRestriction} = 1) = F(\text{Hedging}, \text{Hedging} \times \text{ZScore}, \text{Hedging} \times \text{M/B}, \text{Controls}).$$

4.3 Investment model

$$\text{Investment/LaggedAssets} = \alpha + \beta \text{Hedging} + \text{Interactions} + \text{Controls} + \varepsilon.$$

4.4 True cost of debt

$$\text{PromisedDebtYield} = \text{CostOfDebt} + \text{YieldEquivalentOfExpectedDefaultLoss}.$$

5 Identification and empirical design

The main concern is endogenous hedging. The paper uses tax convexity as an instrument for hedging, lags firm variables relative to loan origination, controls for rich firm and loan characteristics, and tests theory-motivated heterogeneity using distress and growth-opportunity proxies.

6 Tables

6.1 Table I: Summary Statistics

Read: Hedgers have lower raw loan spreads, fewer capex restrictions, larger firms and loans, and lower cash-flow volatility.

Detailed interpretation: Table I motivates the IV strategy because hedgers differ from nonhedgers.

Economic message: Raw differences suggest hedging is associated with better loan terms, but selection must be addressed.

Table I
Summary Statistics

Summary statistics are based on the sample used in regression analyses. Variables are defined in Appendix B.

Panel A: Full Sample Summary Statistics						
	Obs.	Mean	Std. Dev.	p25	p50	p75
<i>Firm Hedging Information</i>						
IR hedging dummy	2,718	0.356	0.479	0	0	1
IR hedging (for hedgers)	869	0.138	0.124	0.044	0.107	0.201
FX hedging dummy	2,718	0.273	0.446	0	0	1
FX hedging (for hedgers)	602	0.075	0.105	0.011	0.037	0.091
IR/FX hedging dummy	2,718	0.501	0.500	0	1	1
IR/FX hedging (for hedgers)	1,143	0.140	0.142	0.040	0.100	0.196
Net IR/FX hedging	886	0.133	0.131	0.039	0.093	0.185
<i>Firm Characteristics</i>						
Firm's previous loans	2,718	1.547	0.866	1.099	1.609	2.197
Log assets	2,718	6.538	1.617	5.324	6.521	7.579
Profitability	2,718	0.136	0.101	0.089	0.131	0.181
Tangibility	2,718	0.333	0.230	0.151	0.274	0.466
Cash flow volatility	2,711	0.759	1.865	0.076	0.148	0.342
M/B	2,718	0.825	1.039	0.142	0.484	1.015
Leverage	2,718	0.295	0.245	0.126	0.259	0.415
Z-score	2,718	1.861	1.220	1.070	1.835	2.543
Distance-to-default	2,718	2.464	2.079	0.984	1.861	3.308
Asset volatility	2,718	0.452	0.375	0.227	0.374	0.574
Covenant restrictions on CAPEX	2,429	0.373	0.483	0	0	1
Investment/lagged assets	2,449	0.077	0.087	0.030	0.052	0.089
<i>Syndicated Loan Characteristics</i>						
Log loan spread (all-in spread drawn)	2,718	4.939	0.824	4.382	5.165	5.570
Loan spread (all-in spread drawn)	2,718	188.602	157.722	80.000	175.000	262.500
Cost of debt	1,482	131.563	110.803	49.987	99.008	187.276
Log loan size (\$M)	2,718	4.697	1.501	3.807	4.828	5.704
Loan size (\$M)	2,718	291.566	712.596	45	125	300
Log maturity (days)	2,679	6.995	0.720	6.579	7.065	7.496
Maturity (days)	2,679	1,344.697	747.454	720	1,170	1,800
Performance pricing	2,718	0.731	0.444	0	1	1
Term loan	2,718	0.381	0.486	0	0	1
Working capital/ corporate purposes	2,718	0.264	0.441	0	0	1
Refinancing	2,718	0.360	0.480	0	0	1
Acquisitions	2,718	0.250	0.433	0	0	1
Backup line	2,718	0.105	0.307	0	0	0
Other	2,718	0.027	0.163	0	0	0
<i>Macro Controls</i>						
Credit spreads	2,718	0.841	0.245	0.630	0.790	0.910
Term spreads	2,718	0.913	0.985	0.210	0.590	1.610

(continued)

(a) Table I Panel A

Table I—Continued

Panel B: Nonhedgers vs. Hedgers			
Variable	(1) Nonhedgers	(2) Hedgers	(3) Diff.: (1) – (2)
Loan spread	201.724	170.521	31.203***
Covenant restrictions on CAPEX	0.413	0.329	0.084***
Investment/lagged assets	0.074	0.081	–0.007**
Log assets	6.058	7.174	–1.116***
Tangibility	0.326	0.344	–0.017*
M/B	0.819	0.826	–0.006
Leverage	0.274	0.323	–0.049***
Z-score	1.825	1.919	–0.094***
Cash flow volatility	0.897	0.560	0.337***
Asset volatility	0.491	0.398	0.092***
Log loan size (\$M)	4.312	5.211	–0.900***
Log maturity (days)	6.992	6.997	–0.004
Cost of debt	137.980	126.093	11.887**

***, **, and * denote statistical significance of the *t*-tests at the 1%, 5%, and 10% level, respectively.

(b) Table I Panel B

6.2 Tables II and III: Loan pricing and capex covenants

Read: Table II shows hedging lowers loan spreads. Table III shows hedging lowers the probability of capex restrictions.

Detailed interpretation: These are the core financial-contracting tables: creditors reward hedging through both price and covenant terms.

Economic message: Hedging improves access to external finance.

Table II
Hedging and Loan Spreads

The dependent variable is log of loan spread. In column (1), results are estimated via OLS. In columns (2) to (4), results are obtained from IV estimations. The instrumental variable for hedging is a firm's tax convexity (Graham and Smith (1999)). Shea's (1997) partial R^2 is a measure of IV relevance. First-stage F -test is the test of excluded IV in the first-stage regression. IR/FX hedging is the total amount of interest rate and currency hedging scaled by assets. All variable definitions are reported in Appendix B. Heteroskedasticity consistent standard errors clustered at the firm level are reported in parentheses.

	(1)	(2)	(3)	(4)
IR/FX hedging	-1.639*** (0.375)	-2.021*** (0.315)	-2.028*** (0.343)	-1.743*** (0.281)
IR/FX hedging $\times$ Z-score			0.496*** (0.165)	
IR/FX hedging $\times$ M/B				-0.407*** (0.114)
Firm's previous loans	-0.067** (0.033)	-0.056* (0.030)	-0.052* (0.031)	-0.051** (0.024)
Log asset	-0.132* (0.077)	-0.133* (0.076)	-0.138** (0.070)	-0.135 (0.084)
Profitability	-0.602** (0.246)	-0.535* (0.297)	-0.573 (0.378)	-0.548** (0.266)
Tangibility	-0.121 (0.094)	-0.122** (0.053)	-0.131 (0.089)	-0.141** (0.068)
Cash flow volatility	0.004 (0.010)	0.004 (0.014)	0.003 (0.007)	0.002 (0.008)
M/B	-0.121*** (0.038)	-0.124*** (0.039)	-0.129*** (0.030)	-0.081** (0.038)
Leverage	0.342* (0.183)	0.363** (0.178)	0.318* (0.174)	0.334 (0.267)
Z-score	-0.105** (0.049)	-0.098*** (0.032)	-0.099** (0.045)	-0.107*** (0.022)
Asset volatility	0.263*** (0.078)	0.374*** (0.101)	0.330*** (0.089)	0.373*** (0.093)
Log loan size	-0.062 (0.046)	-0.057 (0.036)	-0.065* (0.038)	-0.060 (0.037)
Log maturity (in days)	0.132 (0.084)	0.167 (0.123)	0.176* (0.099)	0.172 (0.244)
Credit spread	0.382* (0.218)	0.471* (0.260)	0.442** (0.174)	0.497** (0.235)
Term spread	-0.039 (0.032)	-0.042 (0.036)	-0.048 (0.033)	-0.044 (0.035)
Credit rating dummies	Yes	Yes	Yes	Yes
Performance pricing	Yes	Yes	Yes	Yes
Loan type	Yes	Yes	Yes	Yes
Loan purpose	Yes	Yes	Yes	Yes
Industry effect	Yes	Yes	Yes	Yes
Observations	2,464	2,267	2,267	2,267
Shea's partial R^2 (hedging term)	n.a.	0.129	0.138	0.131
Shea's partial R^2 (interaction term)	n.a.	n.a.	0.149	0.173
First-stage F -test (p -value)	n.a.	0.000	0.000	0.000
Adjusted- R^2 (second-stage)	0.512	0.576	0.578	0.581

*, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

(a) Table II

Table III
Hedging and Capital Expenditure Covenant Restrictions

The dependent variable is a dummy that is equal to one if there is covenant restriction on capital expenditure, otherwise it is zero. The results are obtained from probit-IV estimations. The instrument for hedging is a firm's tax convexity. IR/FX hedging is the total amount of interest rate and currency hedging scaled by assets. All variable definitions are reported in Appendix B. The table reports the marginal effects that are evaluated at the means of continuous independent variables. Heteroskedasticity consistent standard errors clustered at the firm level are reported in parentheses.

	(1)	(2)	(3)
IR/FX hedging	-1.471*** (0.278)	-1.738*** (0.307)	-1.507*** (0.300)
IR/FX hedging $\times$ Z-score		0.603*** (0.123)	
IR/FX hedging $\times$ M/B			-0.362*** (0.071)
Firm's previous loans	-0.022 (0.014)	-0.027** (0.012)	-0.026* (0.014)
Log asset	-0.209** (0.095)	-0.216** (0.096)	-0.212** (0.091)
Profitability	-0.206* (0.119)	-0.223 (0.179)	-0.211** (0.103)
Tangibility	-0.104* (0.058)	-0.106** (0.051)	-0.098 (0.067)
Cash flow volatility	0.003 (0.004)	0.003 (0.004)	0.004 (0.007)
M/B	-0.104*** (0.027)	-0.110*** (0.030)	-0.113*** (0.031)
Leverage	0.143* (0.085)	0.243** (0.115)	0.154* (0.080)
Z-score	-0.135*** (0.038)	-0.127** (0.050)	-0.132*** (0.044)
Asset volatility	0.195** (0.090)	0.175** (0.085)	0.194** (0.086)
Log loan size	0.025* (0.013)	0.035* (0.020)	0.034** (0.014)
Log maturity (in days)	0.183* (0.098)	0.181* (0.097)	0.184** (0.088)
Credit spread	0.223** (0.104)	0.197* (0.113)	0.189 (0.146)
Term spread	-0.038* (0.022)	-0.017 (0.014)	-0.027 (0.023)
Credit rating dummies	Yes	Yes	Yes
Performance pricing	Yes	Yes	Yes
Loan type	Yes	Yes	Yes
Loan purpose	Yes	Yes	Yes
Industry effect	Yes	Yes	Yes
Observations	2,235	2,235	2,235
Pseudo- R^2	0.253	0.258	0.257

*, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

(b) Table III

6.3 Tables IV and VI: Investment and true cost of debt

Read: Table IV links hedging to higher investment. Table VI shows hedging lowers estimated true cost of debt.

Detailed interpretation: These tables connect financing improvements to real outcomes and debt-cost fundamentals.

Economic message: Hedging affects real activity through the credit channel.

Table IV
Investment Equations

The dependent variable is capital expenditure divided by lagged total assets. The results are obtained from IV estimations. The instrument for hedging is a firm's tax convexity. IR/FX hedging is the total amount of interest rate and currency hedging scaled by assets. All variable definitions are reported in Appendix B. Heteroskedasticity consistent standard errors clustered at the firm level are reported in parentheses.

	(1)	(2)	(3)
IR/FX hedging	0.0711*** (0.0201)	0.0652*** (0.0250)	0.0782*** (0.0240)
IR/FX hedging $\times$ Z-score		-0.0223*** (0.0041)	
IR/FX hedging $\times$ M/B			0.0230*** (0.0053)
Firm's previous loans	0.0021 (0.0022)	0.0033** (0.0014)	0.0022* (0.0011)
Log asset	0.0011 (0.0007)	0.0008* (0.0005)	0.0010** (0.0005)
Profitability	0.0010* (0.0005)	0.0006 (0.0004)	0.0011** (0.0005)
Cash flow volatility	-0.0016 (0.0027)	-0.0017 (0.0021)	-0.0015 (0.0026)
M/B	0.0262*** (0.0100)	0.0170** (0.0081)	0.0268*** (0.0091)
Leverage	-0.0025 (0.0025)	-0.0027 (0.0018)	-0.0031** (0.0015)
Z-score	0.0047** (0.0021)	0.0060*** (0.0005)	0.0050** (0.0023)
Asset volatility	-0.0121** (0.0056)	-0.0106** (0.0051)	-0.0105** (0.0050)
Log loan size	0.0015 (0.0010)	0.0014** (0.0006)	0.0016 (0.0013)
Log maturity (in days)	0.0035** (0.0014)	0.0035 (0.0024)	0.0031* (0.0018)
Credit spread	-0.0191* (0.0111)	-0.0161 (0.0189)	-0.0286** (0.0133)
Term spread	0.0061 (0.0043)	0.0068* (0.0038)	0.0051 (0.0043)
Credit rating dummies	Yes	Yes	Yes
Performance pricing	Yes	Yes	Yes
Loan type	Yes	Yes	Yes
Loan purpose	Yes	Yes	Yes
Industry effect	Yes	Yes	Yes
Observations	2,267	2,267	2,267
Adjusted- R^2	0.332	0.334	0.331

*, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

(a) Table IV

Table VI
Hedging and the Cost of Debt

The dependent variable is log of cost of debt (Cooper and Davydenko (2007)). The results are obtained from IV estimations. The instrument for hedging is a firm's tax convexity. IR/FX hedging is the total amount of interest rate and currency hedging scaled by assets. In this table, we calculate the cost of equity, which is an input parameter for computing the cost of debt, using the beta provided by CRSP for firms listed on New York Stock Exchange and American Stock Exchange. All variable definitions are reported in Appendix B. Heteroskedasticity consistent standard errors clustered at the firm level are reported in parentheses.

	(1)	(2)	(3)
IR/FX hedging	-1.920*** (0.305)	-1.960*** (0.324)	-1.706*** (0.264)
IR/FX hedging $\times$ Z-score		0.476*** (0.152)	
IR/FX hedging $\times$ M/B			-0.394*** (0.110)
Firm's previous loans	-0.044* (0.026)	-0.042* (0.025)	-0.040** (0.017)
Log asset	-0.127 (0.078)	-0.131** (0.060)	-0.124 (0.086)
Profitability	-0.493 (0.321)	-0.532 (0.397)	-0.512** (0.246)
Tangibility	-0.106** (0.048)	-0.116* (0.066)	-0.111* (0.062)
Cash flow volatility	0.003 (0.015)	0.003 (0.003)	0.004 (0.009)
M/B	-0.098*** (0.029)	-0.101*** (0.026)	-0.064* (0.038)
Leverage	0.348** (0.161)	0.250 (0.153)	0.318 (0.249)
Z-score	-0.090*** (0.031)	-0.090** (0.043)	-0.096*** (0.026)
Asset volatility	0.329*** (0.082)	0.322*** (0.072)	0.355*** (0.076)
Log loan size	-0.053 (0.035)	-0.061* (0.036)	-0.060* (0.036)
Log maturity (in days)	0.135 (0.120)	0.150* (0.079)	0.162 (0.198)
Credit spread	0.379 (0.243)	0.332 (0.256)	0.428** (0.215)
Term spread	-0.035 (0.029)	-0.04 (0.028)	-0.036 (0.027)
Credit rating dummies	Yes	Yes	Yes
Performance pricing	Yes	Yes	Yes
Loan type	Yes	Yes	Yes
Loan purpose	Yes	Yes	Yes
Industry effect	Yes	Yes	Yes
Observations	1,455	1,455	1,455
Adjusted- R^2	0.484	0.486	0.489

*, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

(b) Table VI

6.4 Table V: Robustness Tests

Read: The main loan spread result survives net hedging, distance-to-default, firm fixed effects, median regression, and timing restrictions.

Detailed interpretation: Table V addresses alternative explanations and measurement concerns.

Economic message: The loan-pricing effect is robust.

7 Result map

- Hedging lowers loan spreads.
- Hedging matters more for financially distressed and high-growth firms.
- Hedging reduces investment restrictions.
- Hedging increases capital expenditure.
- Hedging lowers true cost of debt and not only promised loan spreads.

Table V
Robustness Tests

The dependent variable is log of loan spread. In columns (1) and (2), the estimations are via OLS. Column (3) is a firm fixed effect estimation. Column (4) is a median regression with pseudo- R^2 reported. Column (5) is the deal-level OLS regression. Net IR/FX hedging is net interest rate and net currency hedging scaled by assets. IR/FX hedging is the total amount of interest rate and currency hedging scaled by assets. All variable definitions are reported in Appendix B. Heteroskedasticity consistent standard errors clustered at the firm level are reported in parentheses.

	(1)	(2)	(3)	(4)	(5)
Net IR/FX hedging	-1.737*** (0.385)				
IR/FX hedging		-1.542*** (0.269)	-1.177*** (0.394)	-1.471*** (0.544)	-1.171*** (0.372)
Firm's previous loans	-0.059* (0.032)	-0.073** (0.033)	-0.051** (0.025)	-0.070** (0.031)	-0.078** (0.033)
Log asset	-0.127* (0.073)	-0.139** (0.064)	-0.108* (0.060)	-0.157* (0.093)	-0.150** (0.067)
Profitability	-0.520** (0.227)	-0.692** (0.297)	-0.401* (0.218)	-0.692** (0.304)	-0.559* (0.306)
Tangibility	-0.112 (0.104)	-0.137 (0.091)	-0.105 (0.117)	-0.142* (0.081)	-0.150* (0.089)
Cash flow volatility	0.001 (0.010)	0.002 (0.014)	0.001 (0.011)	0.002 (0.007)	0.003 (0.007)
M/B	-0.103** (0.046)	-0.110*** (0.035)	-0.117** (0.050)	-0.125*** (0.044)	-0.142** (0.056)
Leverage	0.302 (0.196)	0.383** (0.177)	0.257 (0.173)	0.345* (0.200)	0.327* (0.183)
Z-score	-0.124** (0.050)		-0.096** (0.043)	-0.126** (0.056)	-0.111** (0.050)
Distance-to-default		-0.076*** (0.022)			
Asset volatility	0.210*** (0.042)	0.102 (0.247)	0.262** (0.119)	0.225*** (0.036)	0.212*** (0.038)
Log loan size	-0.062 (0.044)	-0.056 (0.039)	-0.03 (0.027)	-0.063 (0.053)	-0.066 (0.053)
Log maturity (in days)	0.144 (0.098)	0.154 (0.101)	0.118 (0.100)	0.165 (0.124)	0.194 (0.166)
Credit spread	0.377** (0.177)	0.384** (0.183)	0.450** (0.184)	0.374* (0.218)	0.381* (0.213)
Term spread	-0.042 (0.031)	-0.075 (0.049)	-0.036 (0.049)	-0.043 (0.034)	-0.057 (0.040)
Credit rating dummies	Yes	Yes	No	Yes	Yes
Performance pricing	Yes	Yes	Yes	Yes	Yes
Loan type	Yes	Yes	Yes	Yes	Yes
Loan purpose	Yes	Yes	Yes	Yes	Yes
Industry effect	Yes	Yes	No	Yes	Yes
Firm effect	No	No	Yes	No	No
Observations	2,211	2,464	2,038	2,464	1,860
Adjusted- R^2	0.554	0.56	0.522	0.37	0.576

8 Conclusion

Campello, Lin, Ma, and Zou show that corporate hedging affects creditors' pricing and contracting decisions, and those financial effects translate into real investment. The incremental contribution is to identify exact mechanisms through which hedging affects corporate outcomes: lower borrowing costs, looser investment covenants, lower true debt costs, and higher investment.